Customer Behaviour Analysis Report

Overview

This report presents a customer behaviour analysis conducted on 500 customers of the Blyss wellness platform. The analysis covers exploratory data analysis, sentiment analysis of customer reviews, K-Means clustering segmentation, and targeted business recommendations for each identified segment.

Customers Analysed	Positive Reviews	Customer Segments	High-Value Customers
500	59.2%	3	169

1. Exploratory Data Analysis

Initial exploratory analysis was performed to understand the distribution of key customer features including booking frequency, average spending, recency of last activity, and service preferences. The data reveals a right-skewed booking distribution with most customers booking between 2 and 7 times, and average spending concentrated in the AUD 100–300 range.

A strong positive correlation ($r = 0.78$) was found between booking frequency and average spending, indicating that more engaged customers also spend more. Days Since Activity shows a negative correlation with both frequency (−0.36) and spending (−0.46), confirming that recency is a reliable proxy for customer value.

Key EDA Findings

Feature Pair	Correlation	Interpretation
Booking Frequency ↔ Avg Spending	0.78	Strong positive relationship
Avg Spending ↔ Days Since Activity	−0.46	Less recent = lower spend
Booking Frequency ↔ Days Since Activity	−0.36	Less recent = fewer bookings

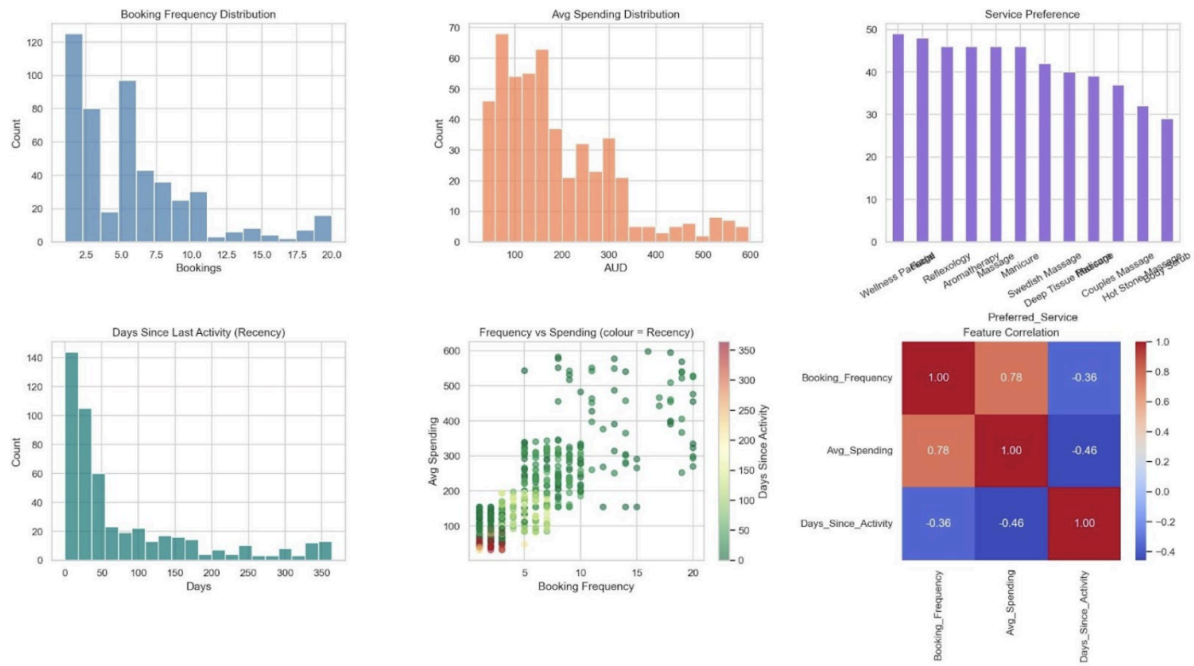

Fig: EDA Overview: Booking Frequency, Spending, Service Preference, Recency Distribution, Frequency vs Spending scatter, and Feature Correlation Heatmap

2. Sentiment Analysis

TextBlob NLP was applied to customer review text to compute a polarity score for each review on a continuous scale of −1 to 1. Reviews were then classified into three sentiment categories based on fixed thresholds.

Classification Thresholds

Label	Condition	Interpretation
Positive	polarity > 0.1	Customer satisfied with the service
Neutral	−0.1 ≤ score ≤ 0.1	Mixed or indifferent experience
Negative	polarity < −0.1	Customer dissatisfied — flag for follow-up

Results

Metric	Value	Note
Mean polarity score	0.283	Overall positive sentiment
Positive reviews	59.2%	Majority satisfied
Neutral reviews	21.6%	Opportunity for improvement
Negative reviews	19.2%	Actionable churn risk signal

The polarity distribution shows a bimodal pattern, with peaks near 0.0 and 1.0. This suggests customers tend to leave either very positive or neutral reviews, with a meaningful tail of negative experiences. The 19.2% negative rate represents an actionable target for service quality improvement.

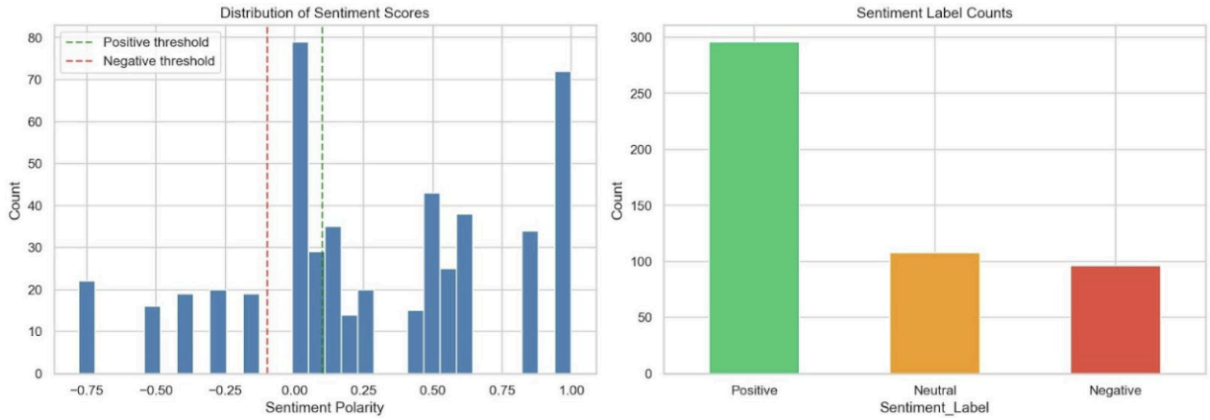

Fig: Left: Distribution of raw sentiment polarity scores. Right: Count of reviews by sentiment label.

3. Customer Segmentation

K-Means clustering was applied to group customers into behavioural segments. StandardScaler normalisation was applied before clustering to ensure no single feature dominates the distance calculations due to scale differences.

Features Used for Clustering

Feature	Description	Rationale
Booking Frequency	Number of sessions booked	Engagement level indicator
Avg Spending	Average spend per booking (AUD)	Revenue contribution
Days Since Activity	Days since last booking	Recency and churn risk signal
Sentiment Score	Mean TextBlob review polarity	Customer satisfaction proxy

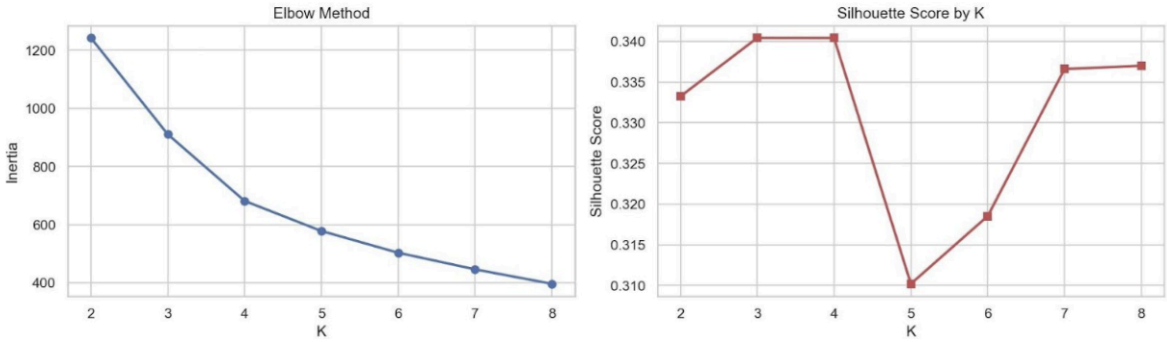

Fig: Left: Elbow Method showing inertia decay. Right: Silhouette Score by K — peak at K = 3 and K = 4 (K = 3 selected).

Selecting the Optimal Number of Clusters

Both the Elbow Method and Silhouette Score were computed for K = 2 through 8. The silhouette score peaks at K = 3 (score = 0.340), while the elbow curve shows a clear inflection between K = 3 and K = 4. K = 3 was selected as it produces the most interpretable and actionable business segments.

PCA was applied for 2-D visualisation, with PC1 explaining 57.4% and PC2 explaining 20.2% of the total variance. The three clusters are well-separated in the PCA projection, confirming meaningful structure in the data.

4. Cluster Profiles

PCA was applied to reduce the four-dimensional feature space to two principal components (PC1 explains 57.4%, PC2 explains 20.2% of variance) for visualisation. The three clusters are well-separated in the PCA projection, confirming meaningful structure in the data.

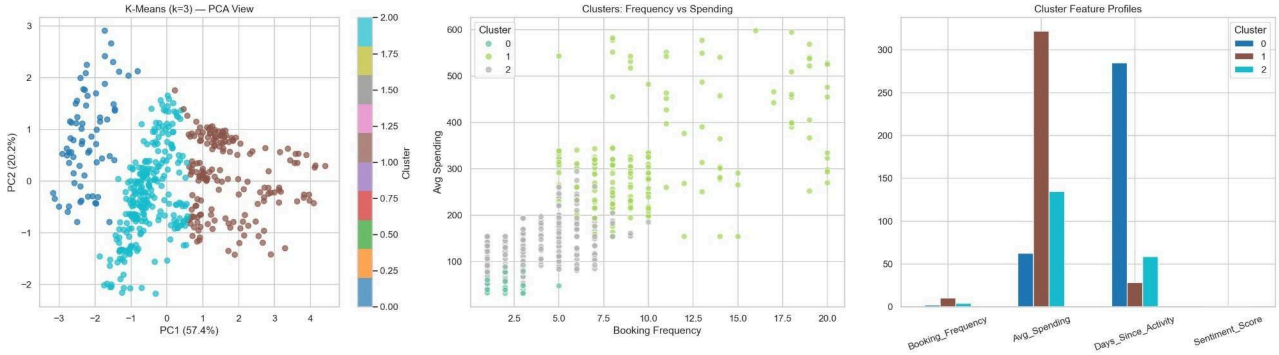

Fig: Left: K-Means clusters in PCA space. Centre: Frequency vs Spending by cluster. Right: Mean feature profile per cluster.

Cluster Summary Table

Cluster	Size	Avg Freq	Avg Spend	Avg Recency	Avg Sentiment	Segment
0	69	1.99	$63	285 days	0.040	At-Risk / Churn
1	169	10.40	$322	28 days	0.600	High-Value
2	262	3.88	$135	59 days	0.140	Occasional

Cluster 0 — At-Risk / Churn (n = 69)

69 customers with very low booking frequency (avg 1.99), minimal average spending ($63), and high inactivity at 285 days since last activity. This group also shows the lowest sentiment score (0.040), indicating dissatisfaction may be contributing to disengagement. These customers represent the highest churn risk and require immediate re-engagement action.

Cluster 1 — High-Value (n = 169)

169 customers form the platform's core revenue base. They book frequently (avg 10.40 sessions), spend significantly ($322 average), have been active very recently (28 days), and leave the most positive reviews (sentiment 0.600). Retaining this segment is the highest-priority business objective.

Cluster 2 — Occasional (n = 262)

The largest segment at 262 customers, showing moderate engagement. They book 3–4 times on average, spend $135 per visit, and have been active within approximately two months. This group represents the strongest upsell opportunity given its size and existing platform familiarity.

5. Insights & Recommendations

Three targeted engagement strategies are recommended, one per customer segment. Priorities are ordered by expected revenue impact.

Priority 1 — High-Value Retention (Cluster 1, 169 customers)

- Introduce a loyalty programme: reward milestone booking (e.g. after every 5th session)
- Provide early and exclusive access to new service launches and therapist availability
- Assign a dedicated therapist or account manager to deepen the customer relationship
- Send personalised thank-you messages post-service with a tailored next-booking offer

Priority 2 — Occasional Upsell (Cluster 2, 262 customers)

- Bundle promotions: offer 3 sessions for the price of 2 to increase booking cadence

- Targeted email campaigns based on each customer's preferred service category
- Seasonal promotions timed around high-demand periods such as public holidays
- Referral scheme: occasional customers are likely to have untapped networks

Priority 3 — At-Risk Re-Engagement (Cluster 0, 69 customers)

- Deploy a win-back campaign with a 15–20% discount voucher
- SMS or push notifications triggered within two weeks of inactivity threshold
- Offer flexible rescheduling with a refund-friendly cancellation policy
- Short feedback survey to understand reasons for disengagement

Strategy Summary

Segment	Size	Goal	Key Tactic	Expected Outcome
High-Value	169	Retention	Loyalty programme + early access	Reduce churn from top revenue tier
Occasional	262	Upsell	Bundle offers + seasonal promotions	Increase booking frequency
At-Risk	69	Re-engagement	Win-back discount + notifications	Recover lapsed revenue